

JavaScript Array HOFs

Higher Order Functions Practice Set • 15 Questions • Easy → Hard

📦 Level 1 — Understand the Core

Q1. `forEach()` — Print Elements

Given an array of numbers, use `forEach()` to print every element one by one.

Hint: `forEach()` ka main purpose yahan transformation nahi, har element par kuch perform karna hai.

Q2. `map()` — Double Values

Given:

```
let arr = [1, 2, 3, 4, 5];
```

`map()` ka use karke ek new array banao jisme har number ka double ho.

Expected:

```
[2, 4, 6, 8, 10]
```

Hint: `map()` → har element ko transform karta hai.

Q3. `filter()` — Even Numbers

Given:

```
let arr = [1, 2, 3, 4, 5, 6, 7, 8];
```

`filter()` ka use karke sirf even numbers ka new array banao.

Expected:

```
[2, 4, 6, 8]
```

Hint: Yahan elements ko select karna hai, transform nahi.

Q4. `reduce()` — Sum

Given:

```
let arr = [10, 20, 30, 40];
```

`reduce()` ka use karke array ka total sum find karo.

Expected: 100

Hint: Multiple values ko single result me combine karna hai.

Q5. `for...of` — Print Values

Given an array of names, `for...of` ka use karke har name print karo.

Hint: `for...of` directly array ke values deta hai.

📦 Level 2 — Individual HOF Mastery

Q6. `map()` — Squares

Given:

```
let arr = [1, 2, 3, 4, 5];
```

`map()` ka use karke har element ka square return karo.

Expected:

```
[1, 4, 9, 16, 25]
```

Q7. `filter()` — Greater Than 50

Given:

```
let arr = [20, 55, 10, 80, 45, 90];
```

`filter()` ka use karke sirf woh numbers return karo jo `50` se greater hain.

Expected:

```
[55, 80, 90]
```

Q8. `reduce()` — Find Maximum

Given:

```
let arr = [10, 45, 23, 89, 34];
```

`reduce()` ka use karke largest number find karo.

Expected: 89

⚠️ `Math.max()` use nahi karna.

Q9. `find()` — First Matching Element

Given:

```
let arr = [10, 25, 35, 60, 80];
```

`find()` ka use karke first number greater than 30 find karo.

Expected: 35

Important: `find()` ko `filter()` se differentiate karna hai — hume sirf first match chahiye.

Q10. `some()` vs `every()`

Given:

```
let arr = [20, 40, 60, 80];
```

1. Check karo kya at least one number `50` se greater hai.
2. Check karo kya all numbers `10` se greater hain.

Hint: Dono cases me alag HOF naturally fit hota hai.

□ Level 3 — Combination

Q11. `filter()` + `map()` □

Given:

```
let arr = [1, 2, 3, 4, 5, 6];
```

Pehle sirf even numbers select karo, phir unke squares nikalo.

Expected:

```
[4, 16, 36]
```

Hint: `filter()` → selection, `map()` → transformation.

Q12. `filter()` + `reduce()` □

Given:

```
let arr = [10, 15, 20, 25, 30];
```

Sirf even numbers ka sum calculate karo.

Expected: 60

Hint: Pehle decide karo kaunse elements chahiye, phir unka single result nikalo.

Q13. Objects + `filter()` + `map()`

Given:

```
let users = [  
  { name: "Aman", age: 17 },  
  { name: "Kartik", age: 20 },  
  { name: "Rohit", age: 22 },  
  { name: "Rahul", age: 16 }  
];
```

18 ya usse zyada age wale users ke names ka new array banao.

Expected:

```
["Kartik", "Rohit"]
```

Hint: Pehle users select karo, phir unse names extract karo.

Level 4 — Real Problem Solving

Q14. Product Data Challenge

Given:

```
let products = [  
  { name: "Laptop", price: 60000, category: "electronics" },  
  { name: "Phone", price: 30000, category: "electronics" },  
  { name: "Shoes", price: 3000, category: "fashion" },  
  { name: "Watch", price: 5000, category: "fashion" }  
];
```

Sirf `"electronics"` category ke products select karo aur unke names ka new array banao.

Expected:

```
["Laptop", "Phone"]
```

Hint: Yahan objects + filter + map combine honge.

Q15. 🦴 Final HOF Challenge

Given:

```
let students = [  
  { name: "Aman", marks: 72 },  
  { name: "Kartik", marks: 91 },  
  { name: "Rohit", marks: 85 },  
  { name: "Rahul", marks: 64 },  
  { name: "Arjun", marks: 95 }  
];
```

Program likho jo:

1. Sirf `80+` marks wale students select kare.
2. Unke names ka new array banaye.
3. Check kare ki koi bhi student 95 ya usse zyada marks score karta hai ya nahi.
4. Selected students ke marks ka average calculate kare.

Hint: Isme ek hi HOF enough nahi hoga. Sochna hai ki selection → transformation → checking → accumulation ke liye kaunse methods fit hote hain.